GH3036 SPO2 Performance Report

Version: 1.0

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Preface

Purpose

This document is intended to reflect the SPO2 performance test results of GH3036.

Audience

This document is intended for:

- Systems and product engineers
- Field application engineers
- Test Engineers
- R&D Engineers

Release Notes

This document is the first release for software version Package_V00.02.00.00

Revision History

Version	Date	Description
1.0	2025-11-14	Official release

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1 Config Introduction

Software version: Package_V00.02.00.00, mainly includes:

1. APP: ghealth_ V1.0.9.13_build005
2. Algorithm version:

   GH_SPO2_pre_pv_v2.1.10.0_ 4e61c7
3. Firmware: GR551x_VX.X_DEMO_DRV_LIB_GH3036.bin
4. Driver library: GH(M)3X_DEMO_v0.2.1.0_ALGO.bin

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2 Test Requirements

Table 2-1 Test Requirement Details

Condition	Descriptions
Test environment	In an office; walk-in constant temperature and humidity test chamber
Test environment temperature	25°C, 5°C
Testee gender	Male and female
Testee age	20- 40 years old
Testee BMI	18- 30 kg/m^2
Testee skin color level	Levels 1- 10

Note:

Skin Tone Guide:

The Pantone SkinTone Guide										
Goodix Index	1	2	3	4	5	6	7	8	9	10
Pantone Index	1Y01SP	5Y01SP	2Y06SP	4Y07SP	2Y10SP	3Y11SP	2Y13SP	3R13SP	1R14SP	4R15SP

3 Result

1. Sample quantity: Based on the data quantity recorded in the table.
2. Test method: The testee wears GH3036 device and reference device(Masimo) with a snug but comfortable fit, connects the SPO2 recording App (GHealth) to both the GH3036 device and the reference device. Then perform tests in different scenarios.
3. Test scenarios:
    1. Sit still: SPO2 data have been collected @room temperature. Each test lasts 1 minute for one testee. Repeat the test for 5 times.
    2. Stand still: SPO2 data have been collected @room temperature. Each test lasts 1 minute for one testee. Repeat the test for 5 times.
    3. Low temperature: Start to collect SPO2 data at rest after sitting for 7mins in the low temperature environment (avoid moving). Each test lasts 1 minute for one testee. Repeat the test for 5 times.
    4. Oxygen-deficient/Oxygen-enriched Test Scenario: the testee sits still at room temperature with both hands resting and face wearing an oxygen mask. The blood oxygen levels are maintained in the ranges of 90%–95%, 85%–90%, and 80%–85%. Perform the test at each oxygen level for 1 minute. Repeat the test for 5 times.

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3.1 Test results of yellow skin

Table 3-1 Test results of yellow skin

Test Scene	Sample quantity	Algorithm Version	Output Rate@60s	Output Duration	RMSE	≤±3% Ratio	Fake Low O2 % for 1st value	Fake High O2 % for 1st value
Sitting still	100	Premium	94.00%	12.50	1.29	93.90%	10.00%	/
Standing still	100	Premium	100.00%	11.40	1.33	96.10%	7.00%	/
90-95 Sitting Still	100	Premium	100.00%	13.00	1.37	96.90%	/	1.00%
85-90 Sitting Still	100	Premium	100.00%	13.90	1.86	90.30%	/	2.00%
80-85 Sitting Still	100	Premium	94.00%	13.50	2.58	83.40%	/	3.00%

3.2 Test results of black skin

Table 3-2 Test results of black skin

Test Scene	Sample quantity	Algorithm Version	Output Rate@60s	Output Duration	RMSE	≤±3% Ratio	Fake Low O2 % for 1st value	Fake High O2 % for 1st value
Sitting still	100	Premium	97.00%	12.49	1.62	97.43%	4.00%	/
Standing still	100	Premium	93.00%	13.65	1.62	95.55%	8.00%	/
90-95 Sitting Still	100	Premium	97.00%	12.75	2.59	79.58%	/	5.00%
85-90 Sitting Still	100	Premium	94.00%	14.14	3.84	62.66%	/	13.00%
80-85 Sitting Still	100	Premium	99.00%	14.10	5.04	47.98%	/	6.00%

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4 Legend

4.1 Sitting still

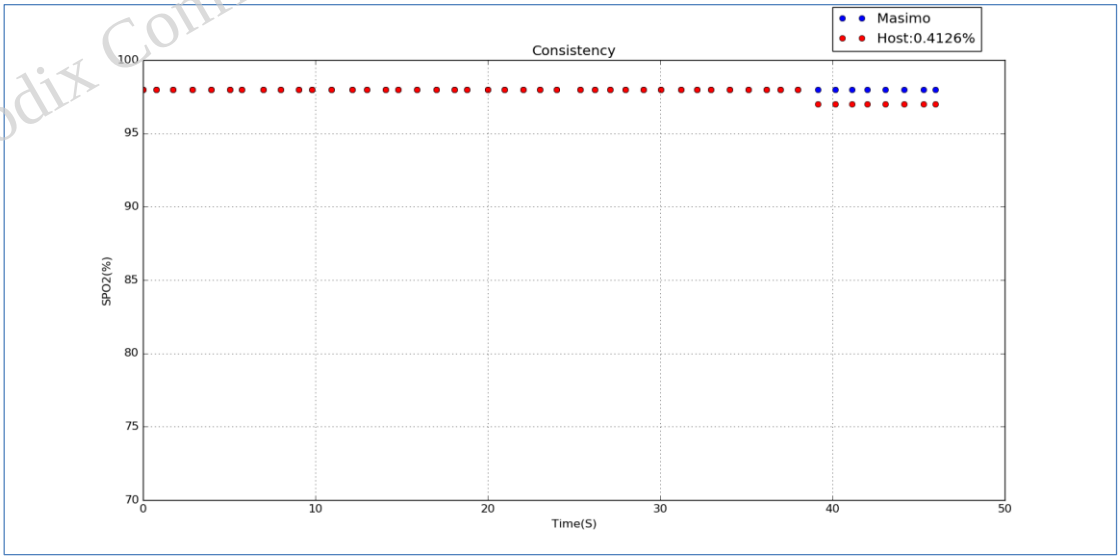

Figure 4-1 Sitting still

4.2 Standing still

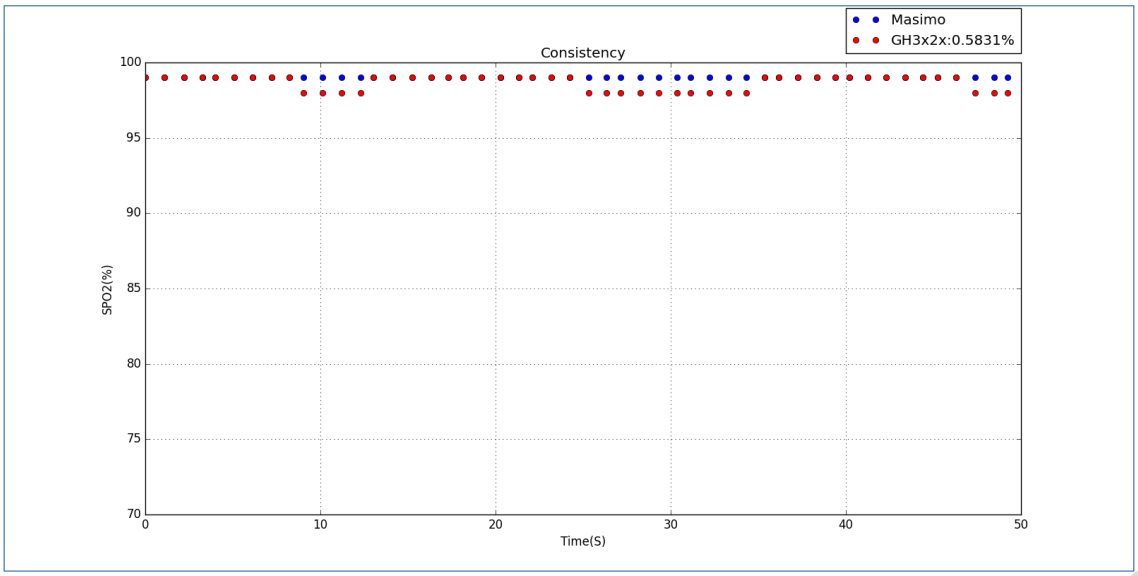

Figure 4-2 Standing still

4.3 Min-level Low O2

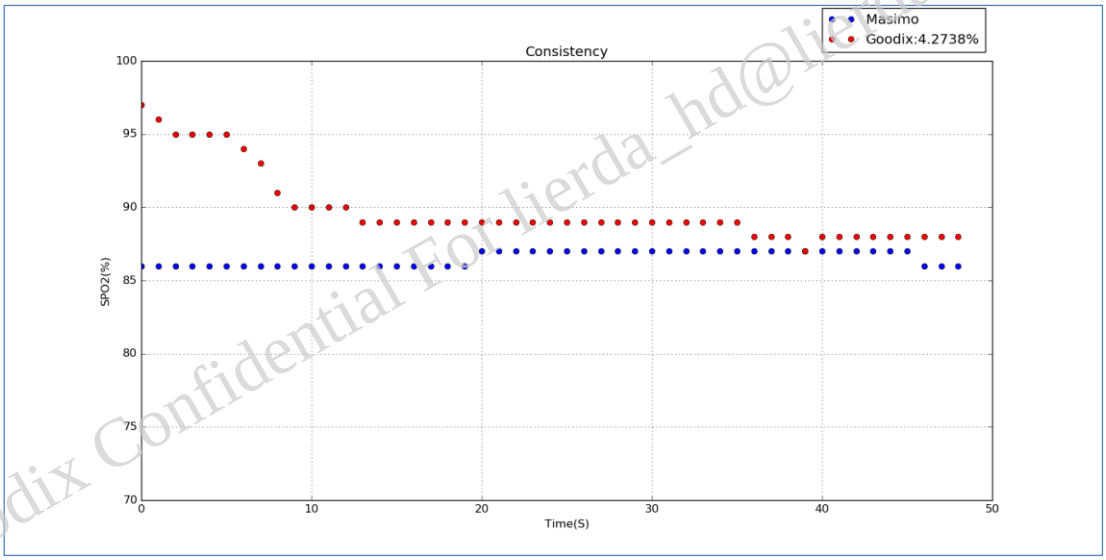

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Figure 4-3 Min-level Low O2

5 Terminology

Table 5 Terminology

Indicator	Definition
Output Rate@30s	Test N times, of which M times can produce values within 30s, then the value rate = M/N*100%.
Output Rate@60s	Test N times, of which M times can produce values within 60s, then the value rate = M/N*100%.
Output Time	Average value time within 1min.
RMSE	Root Mean Squared Error, also known as standard error, is defined as , i=1, 2, 3, ...n. In a limited number of measurements, the root mean square error is often expressed by the following formula: √[∑(di^2)/n], where: n is the number of measurements; di is the deviation between a set of measured values and the true value.
≤±3% Ratio	In the data that produce values within 1min, the blood oxygen deviation between the test sample and the standard equipment is within (inclusive) ±3%.

Indicator	Definition
Fake Low O2 % for 1st value	In hypoxic scenarios, the number of samples with the first value ⩾95% and the difference with the standard equipment >3% accounts for the total number of samples
Fake High O2 % for 1st value	In normoxic scenarios, the number of samples with blood oxygen values <95% and the difference with the standard equipment >3% accounts for the total number of samples
High Confidence %	Data with high confidence level, or data that algorithm considering highly reliability.

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